Adaptive Quantum Error Mitigation Under Fixed Shot Budgets: Operating Regimes, Abstention, and Noise-Model Shift

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Abstract

Quantum error mitigation (QEM) trades bias reduction against auxiliary executions and sampling variance, so a method that helps one circuit may waste resources or become unstable elsewhere. We compare unmitigated estimation, full-assignment-matrix readout mitigation, and linear zero-noise extrapolation (ZNE) under identical total shot budgets. The simulator benchmark combines GHZ and fixed-seed hardware-efficient circuits with widths 2, 4, and 6, depths 1, 3, and 6 where applicable, four main noise levels, three shot budgets, and 50 deterministic repeats. A cost-aware oracle labels 144 operating cells as 103 practical ties, 18 readout winners, 14 ZNE winners, and 9 inconclusive. A frozen, non-ML rule selector reaches only 41.4% decision compatibility on a 29-cell deterministic holdout and 50.0% under four unseen synthetic noise shifts. The benchmark retains 99 explicit readout failures and 179 of 432 method summaries with negative RMSE gain. An additional preregistered ZNE validity gate accepts only 5 of 16 simulated circuit–noise–budget cells; none of the 8 tested hardware-efficient cells passes every check. In a separate factorial study, combined readout mitigation and ZNE has the lowest expectation RMSE in 12 of 16 preregistered conditions, yet all 16 combined-ZNE conditions fail at least one frozen validity criterion. Its near-tie decision-error probability remains between 0.26 and 0.48. In a fixed-budget MaxCut application, lower energy RMSE does not reliably improve top-1 QAOA-candidate selection, and only 2 of 828 completed selections certify a winner under simultaneous intervals. These simulated results support selective mitigation and abstention as research questions, but do not validate a general selector or establish hardware performance.

1 Introduction

QEM can reduce expectation-value bias without logical encoding, but its additional circuits and statistical amplification consume finite resources [10, 6, 3]. Comparing methods at unequal execution cost can therefore manufacture an apparent advantage. Even under a fixed budget, the best method can change with circuit structure, observable, noise, and sampling regime.

This work asks when mitigation improves RMSE at fixed total shots, when methods are practically indistinguishable, whether a simple selector can choose safely without the exact reference, when apparent ZNE improvement is credible, and whether readout correction and ZNE cooperate under one resource ledger. We also ask whether a mitigated energy estimate improves a complete MaxCut/QAOA candidate-selection decision rather than only an expectation metric. The contribution is a reproducible simulator benchmark with repeat-level data, explicit execution ledgers, cost-aware labels, abstention rules, and evaluation under noise-model mismatch. It is not a proposal for new QEM theory or a claim of quantum-hardware advantage.

2 Methodology

2.1 Circuit families and observables

The GHZ family prepares

$$|\mathrm{GHZ}_w\rangle = (|0\rangle^{\otimes w} + |1\rangle^{\otimes w})/\sqrt{2}, \tag{1}$$

with analytic observables $X^{\otimes w} = 1$ and neighboring $ZZ = 1$. The hardware-efficient family alternates fixed-seed single-qubit rotations and nearest-neighbor entanglers at depths 1, 3, and 6. Its Pauli observables and statevector references are fixed before noisy sampling; some references are deliberately away from ± 1 to avoid a stabilizer-only benchmark.

For repeat r, $e_r = \widehat{\mu}_r - \mu_{\mathrm{ref}}$. We report bias, sample variance, and

$$\mathrm{RMSE} = \sqrt{R^{-1}\sum_{r=1}^{R} e_r^2}. \tag{2}$$

Paired RMSE-gain intervals use deterministic bias-corrected and accelerated bootstrap resampling. Decision-error probability is the fraction of repeats whose estimated sign disagrees with the reference decision; Wilson intervals quantify its binomial uncertainty.

2.2 Mitigation and fixed budgets

Raw estimation spends the full budget on the target. Readout mitigation allocates 20% to all 2^w calibration

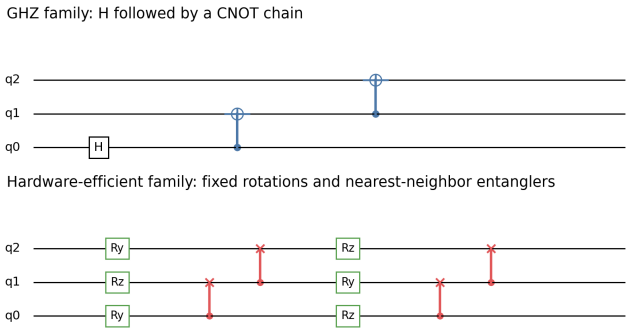


Figure 1: The benchmark combines an analytically controlled GHZ family with a nontrivial hardware-efficient depth grid; both diagrams are conceptual and all executed circuits are generated deterministically in code.

states and 80% to the target. It solves a constrained assignment-matrix correction and reports ill-conditioning explicitly rather than silently inverting an unstable matrix [1]. Linear ZNE splits the same budget over global folds $\lambda \in \{1, 3, 5\}$ and uses the preregistered fit

$$\widehat{\mu}(\lambda) = \beta_0 + \beta_1 \lambda, \qquad \widehat{\mu}_{\mathrm{ZNE}} = \beta_0, \tag{3}$$

without clipping out-of-range extrapolates [4].

2.3 Combined mitigation and factorial interaction

The combined estimator calibrates one full assignment matrix per repeat, corrects the probability distribution at each fold, converts each corrected distribution to an expectation, and then applies the same linear extrapolation. It reserves 20% of the total budget for calibration. The remaining shots are either divided equally across folds (balanced) or allocated 50%/30%/20% to $\lambda = 1, 3, 5$ (base-heavy). Intermediate counts, corrected estimates, extrapolates, and every calibration and folded execution remain in the repeat-level ledger. For paired absolute errors E, the factorial interaction is

$$I = E_{\mathrm{combined}} - E_{\mathrm{readout}} - E_{\mathrm{ZNE}} + E_{\mathrm{raw}}. \tag{4}$$

Negative I indicates that the joint error is below the additive single-method contrast; it does not by itself establish a credible extrapolation mechanism.

2.4 ZNE validity and falsification

An extrapolate is not treated as credible merely because it moves toward the exact reference. A preregistered gate additionally requires retained signal at the largest fold, physical expectation and signed-probability diagnostics, positive paired gain with uncertainty, a positive advantage over matched-cost garbage folding, and no excess of destroyed or unstable repeats. Garbage folding increases execution cost without implementing the intended unitary-equivalent noise scaling and therefore serves only as a negative control. The gate uses 50 deterministic repeats per cell; rejected cells retain every failed criterion and are never converted into ZNE wins.

2.5 Cost-aware labels and frozen selector

For each fully completed cell, the oracle combines RMSE with an execution-overhead penalty. Differences below the preregistered RMSE threshold 0.005 are labeled `practical_tie`; incomplete three-method comparisons are `inconclusive`. A deterministic hash split reserves at least 20% of cells. The selector is then frozen before hold-out outcomes are inspected. It uses only width, depth, declared noise regime, shots, calibration conditioning, and ZNE fit residuals. It can select raw, readout, or ZNE, or abstain; it never receives the statevector reference or oracle label.

2.6 MaxCut/QAOA application decision

For a graph $G = (V, E)$ and bitstring z, the exactly checkable cut value is

$$C_G(z) = \sum_{(u,v)\in E} \mathbf{1}[z_u \neq z_v]. \tag{5}$$

Exhaustive enumeration fixes the classical optimum and all degenerate optimal bitstrings for one width-4 ring, one deterministic width-6 sparse graph, and one deterministic width-8 medium/dense graph. A frozen grid supplies five $p = 1$ QAOA candidates per graph. Noiseless statevector expectations determine the incumbent, challengers, ideal ranks, and clear versus near-tie regimes before noisy data exist. One computational-basis shot yields one complete cut sample; edges do not receive independent free shots.

Each method receives one application-level budget $B \in \{4096, 16384\}$ for selecting among all five candidates. Raw estimation divides B among candidates. Readout charges calibration within B, and ZNE charges all three folded executions within B. We report exact top-1 accuracy, top-k recall, simple regret against ideal candidate energy, energy RMSE, and unresolved probability separately. At nested shot checkpoints, a candidate is certified only if its adjusted uncertainty interval is separated from every challenger under a Bonferroni simultaneous-coverage rule; the runtime selector never receives the ideal winner.

3 Experiment Setup

The main grid crosses widths $2, 4, 6$, hardware-efficient depths $1, 3, 6$, noise levels zero/low/medium/high, shot budgets 1,024/4,096/16,384, three methods, and 50 repeats. GHZ depths 3 and 6 remain explicit `not_applicable` cells. All seeds derive from root seed 20261102. Qiskit Aer supplies ideal statevectors and

Table 1: Most simulated operating cells are practical ties; a single universal mitigation winner is not supported.

Cost-aware label	Cells
Practical tie	103
Readout	18
Linear ZNE	14
Inconclusive	9

finite-shot noisy simulation [8]. The main model combines independent assignment error with one- and two-qubit depolarizing channels. The method comparison follows the broader need for common QEM benchmarks [2]; the scalable-readout track is motivated by assignment-matrix growth and reduced calibration models [7].

A separate 4,096-shot shift set tests intermediate mixed noise, asymmetric readout, coherent over-rotation, and correlated ZZ error. These are controlled synthetic stress tests, not calibrated device models. Every manifest row has a raw row with status `complete`, `failed`, or `not_applicable`; no failed cell is silently dropped.

The combined study preregisters six diagnostic cells: readout-dominated, gate-noise-dominated, practical-tie, ZNE-invalid, 0.02-margin decision, and 0.20-margin decision conditions. It crosses budgets 4,096 and 16,384, both frozen allocation policies, four factorial methods, and 50 paired repeats. Matched-cost garbage folding is used only by the validity gate and is excluded from method ranking.

The application manifest contains three graphs, five candidates, three methods, two budgets, and 50 deterministic repeats, for 4,500 candidate-level rows and 900 selection groups. Candidate construction uses graph structure and ideal statevector values only. The exact classical solver is used for post hoc certification, never to guide the noisy selector.

4 Results

4.1 Operating regimes

Table 1 shows that practical equivalence dominates the grid. Readout and ZNE win distinct subsets, while nine cells are inconclusive because the 50-repeat completeness contract is not met. Figure 2 shows how these labels vary with simulated noise. The map prevents small numerical RMSE differences from being presented as scientific method changes.

Mitigation can also worsen estimation: 179 of 432 method summaries have negative RMSE gain relative to raw estimation. Readout produces 99 explicit hardware-efficient failures, concentrated at width 6, 1,024 shots, and medium/high noise. These failures are part of method performance, not missing data.

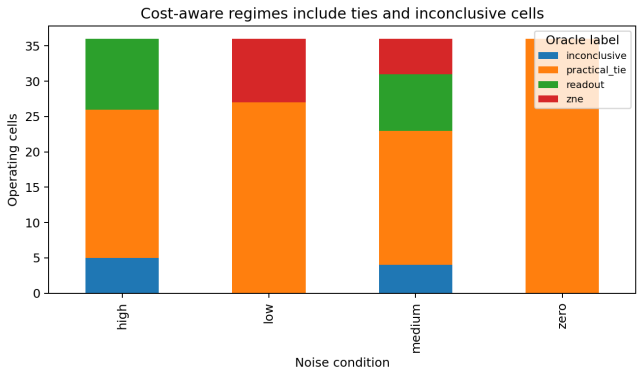


Figure 2: Cost-aware labels retain practical ties and incomplete comparisons instead of forcing every simulated cell to declare a winner.

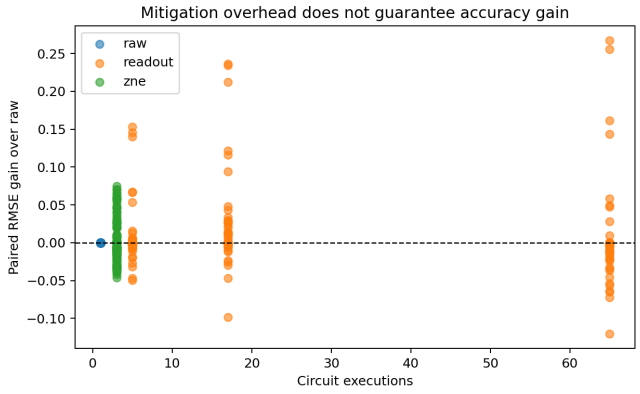


Figure 3: Higher simulated execution overhead does not guarantee positive RMSE gain; negative gains remain visible rather than being clipped or filtered.

4.2 ZNE validity under negative controls

The probability-level diagnostic contains 800 simulated repeat records across 16 preregistered cells. Of these, 197 repeats are classified as extrapolation-unstable and 5 as signal-destroyed. The cell-level validity gate accepts 5 cells and rejects 11. All 8 hardware-efficient cells are rejected; their apparent corrections do not survive the combined gain-uncertainty and trend requirements. The five accepted cells are Bell-state conditions, so this result narrows rather than broadens the evidence for linear ZNE.

4.3 Combined QEM under one budget

Combined mitigation has the lowest expectation RMSE in 12 of 16 preregistered circuit–budget–allocation conditions. This descriptive win count is insufficient for a mechanism claim: combined RMSE gain versus raw ranges from negative to positive, and additional execution burden does not guarantee improvement (Fig. 6).

Exactly 12 of 24 paired factorial contrasts are negative;

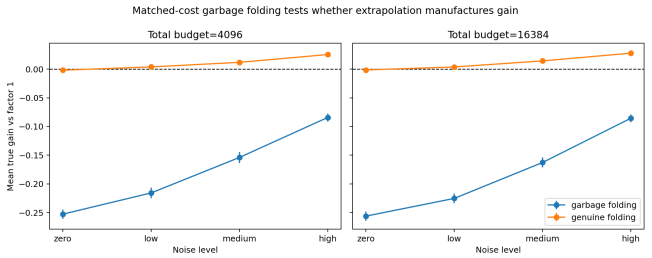


Figure 4: Matched-cost garbage folding separates an apparent correction from a unitary-equivalent ZNE effect; only cells passing the full preregistered gate are reported as credible.

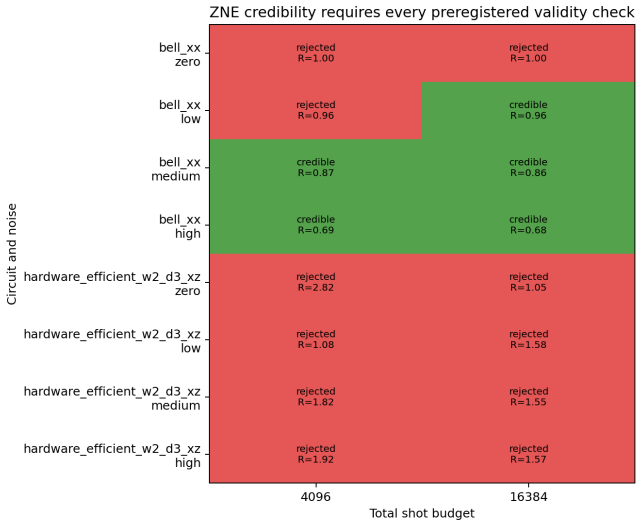


Figure 5: Signal retention and extrapolation stability vary by simulated circuit, noise, and budget; the hardware-efficient cells fail the combined validity gate despite some favorable individual diagnostics.

the remainder are positive. Their uncertainty intervals frequently overlap zero (Fig. 7), so the data do not support a uniform cooperative or antagonistic interaction. More decisively, all 16 of 16 combined-ZNE conditions are rejected by at least one frozen signal, physicality, uncertainty, trend, or garbage-control requirement. The RMSE improvements therefore remain simulated estimator outcomes rather than validated ZNE effects.

4.4 Selector holdout and noise shift

On 29 held-out cells, the frozen selector achieves 41.4% decision compatibility. Mean RMSE regret is 0.0207 and median regret is 0.000397. Strict label agreement is lower because a practical tie can contain several acceptable methods; compatibility is therefore the primary decision metric.

Under the four unseen noise shifts, compatibility is 50.0%, mean regret is 0.00771, and median regret is 0. Negative regret can occur because the cost-aware oracle

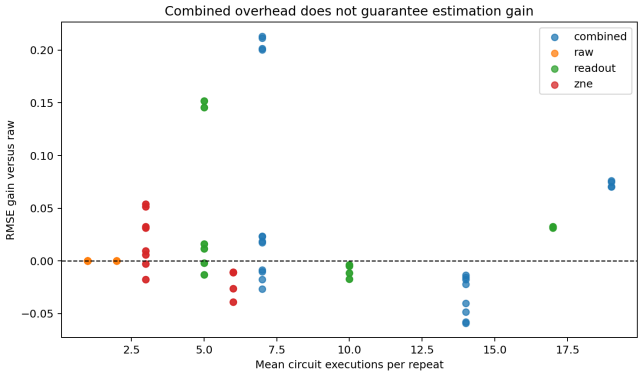


Figure 6: Combined readout correction and ZNE often lowers simulated expectation RMSE, but some conditions remain worse than raw despite greater execution overhead.

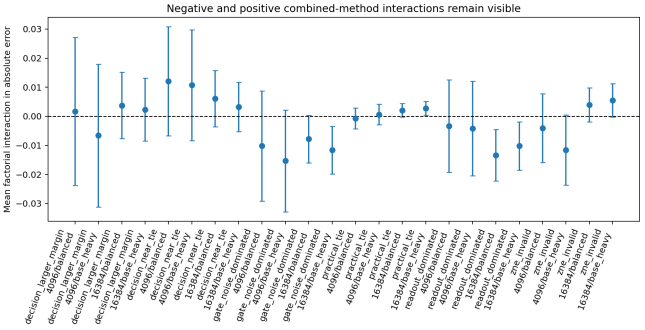


Figure 7: Paired factorial interactions change sign across preregistered cells, budgets, and allocations; many intervals cross zero, rejecting a universal synergy claim.

penalizes overhead while the plotted quantity is RMSE alone; it is not evidence that the selector beats the oracle's declared utility.

4.5 Decision reliability

Expectation RMSE and decision reliability are distinct. Figure 10 reports sign-decision error with Wilson uncertainty. A method may reduce continuous estimation error without reducing the probability of a wrong downstream decision.

The controlled combined-QEM decision study sharpens this separation. At margin 0.02, combined decision-error probability ranges from 0.26 to 0.48 across budgets and allocations and does not consistently beat readout alone. At margin 0.20, nearly all methods resolve the decision. Combined mitigation can therefore improve an expectation estimate while leaving a resource-constrained near-tie unresolved.

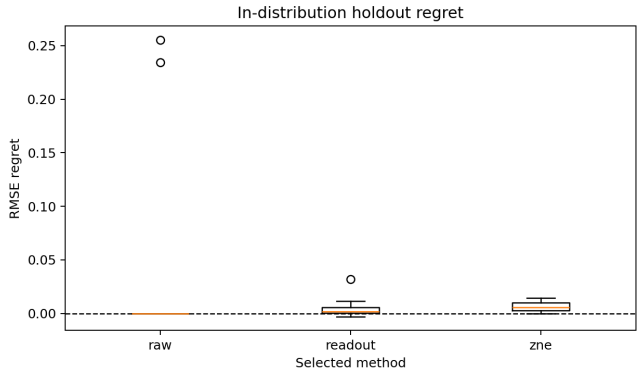


Figure 8: The frozen selector often has small regret but retains substantial positive outliers, so the holdout does not validate a generally reliable policy.

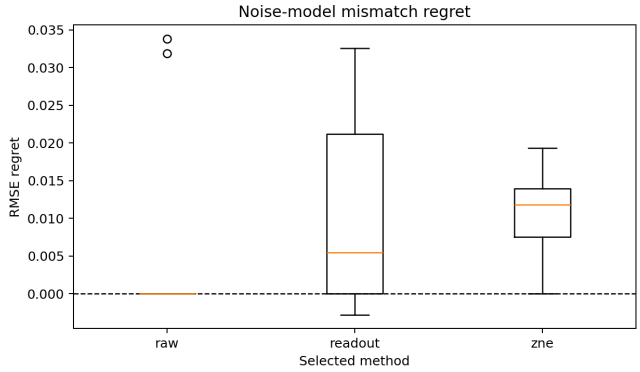


Figure 9: Noise-model mismatch exposes method-dependent regret and outliers; median performance alone would hide these failures.

4.6 Scalable readout correction under model mismatch

The scalable-readout track compares full assignment, tensor-product, observed-subspace, and matrix-free observed-subspace correction. Its final stress test contains 9600 paired simulated rows. All methods share target counts and one fixed total budget; the calibration model remains independent while the target channel is matched independent, asymmetric, pair-correlated, or deliberately mismatched. Bias and repeat sampling spread are reported separately.

In the representative width-6 broad-output, 16,384-shot cell, deliberate calibration mismatch raises the 0.05-margin threshold-decision error to 0.88–0.94 across methods. The matched independent condition remains near zero. Figure 12 therefore shows that classical memory reduction and solver convergence do not establish readout-model validity. Decision risk is evaluated separately from expectation RMSE, consistent with recent cost-aware QEM analysis [9].

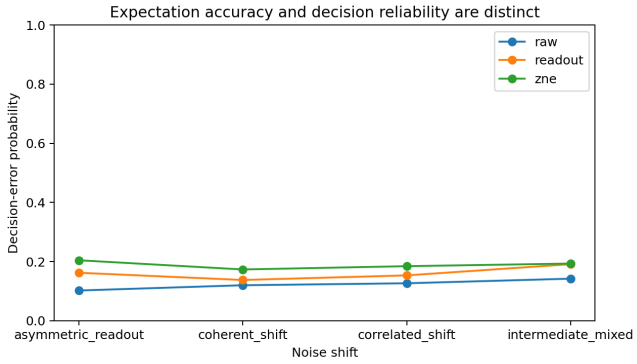


Figure 10: Decision-error probability changes by method and synthetic noise shift, showing that expectation accuracy is not a sufficient proxy for decision reliability.

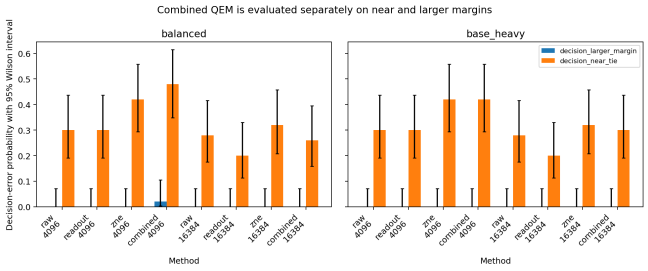


Figure 11: The larger decision margin is resolved, whereas the 0.02 near-tie retains substantial Wilson-bounded error and no consistent combined-QEM advantage.

4.7 MaxCut/QAOA candidate selection

All completed clear-ranking selections identify the ideal best candidate. Near ties separate estimator quality from decision quality (Fig. 13). At 4,096 shots, near-tie error is 0.28 for raw, 0.32 for readout among completed selections, and 0.36 for ZNE; at 16,384 shots it is 0.17, 0.179, and 0.42. On the completed 4,096-shot ring and sparse strata, readout lowers energy RMSE from 0.183 to 0.148 and from 0.228 to 0.191, respectively, yet worsens top-1 selection on both; its medium/dense stratum fails entirely. Across completed repeats, the RMSE–selection-error correlation is only 0.153. Better energy estimation therefore does not imply a better discrete choice in this study.

Simple regret is smaller when the wrong candidate is nearly equivalent, so it and exact top-1 accuracy lead to different emphasis (Fig. 14). At 16,384 shots, mean regret is 0.0130 for raw, 0.0125 for completed readout selections, and 0.0378 for ZNE. Readout fails all 50 medium/dense selections at 4,096 shots and 22 of 50 at 16,384 shots because its assignment correction is ill-conditioned; these 72 failed groups are not treated as abstentions or successes. Results also vary by instance: ZNE's 16,384-shot top-1 error is 0.08 on the ring, 0.34 on the sparse graph, and

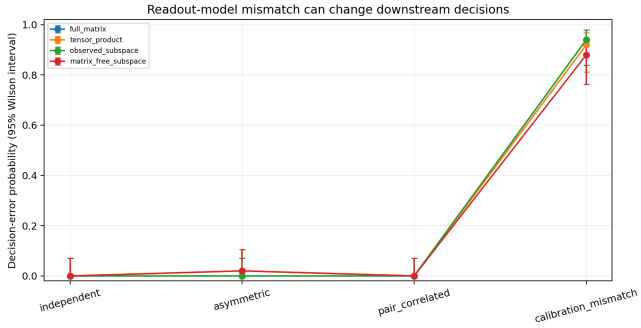


Figure 12: A deliberately mismatched simulated calibration channel causes high 0.05-margin decision error for every correction strategy; reduced storage does not repair model misspecification.

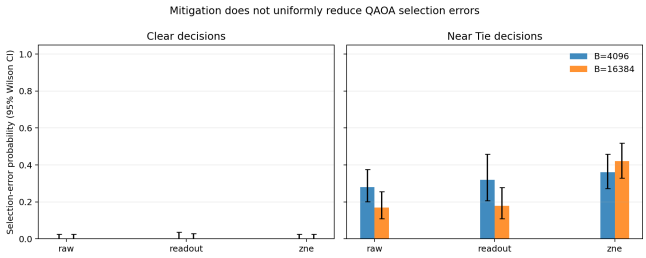


Figure 13: Clear QAOA candidate rankings are selected correctly, whereas near-tie error remains substantial and ZNE worsens when the larger budget is used.

0.50 on the medium/dense graph.

Certification is much stricter than selecting a point-estimate winner. Only two of 828 completed selection groups certify before exhausting the declared budget, both raw width-4 ring cases at 16,384 shots. The remaining 826 completed groups are unresolved, and 72 groups fail explicitly (Fig. 15). QEM overhead therefore buys no scientifically meaningful certification improvement in this frozen application grid.

5 Conclusion

No QEM method is uniformly useful under the declared fixed-budget simulator grid. Most operating cells are practical ties, some favor readout or ZNE, and low-shot readout calibration can fail explicitly. The frozen rule selector is reproducible and explainable, but its modest holdout and shift compatibility make it a baseline for further study rather than a successful adaptive policy. The strongest supported conclusion is operational: mitigation should be compared with raw estimation under equal resource accounting, with ties, negative gains, abstention, and failed ZNE validity checks preserved. In this benchmark, apparent extrapolation gains are more common than gains that survive the full falsification gate. Combining readout correction with ZNE improves RMSE

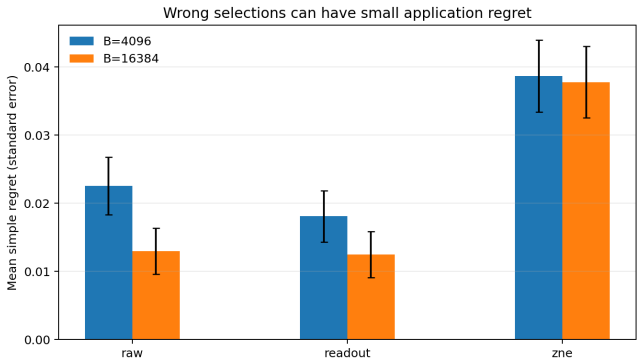


Figure 14: Application regret and exact top-1 error are distinct: a wrong near-tie choice can have low regret, while mitigation overhead does not ensure lower regret.

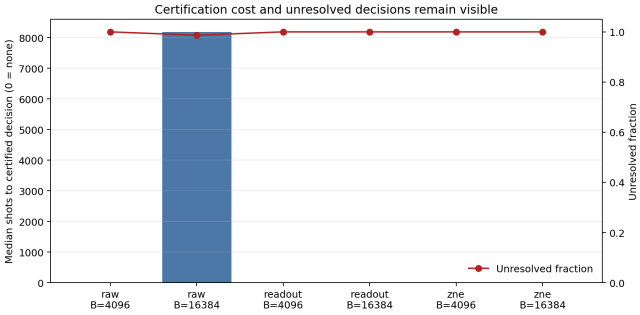


Figure 15: Simultaneous-interval certification is achieved in only two completed selections; nearly every completed method–instance–budget repeat remains unresolved.

in selected simulated conditions, but neither its factorial interaction nor its downstream decision error supports a universal synergy claim, and no combined condition passes the frozen validity gate. The MaxCut application strengthens that boundary: lower energy RMSE is not a reliable proxy for top-1 selection, simple regret, or certified decision, and the added QEM executions do not produce a general application advantage.

6 Discussion

6.1 Supported findings

The data directly support regime dependence, nontrivial execution overhead, explicit readout instability, negative mitigation gains, and a gap between estimation RMSE and decision reliability. The deterministic holdout also shows that intuitive rules do not generalize reliably across the present circuit and noise grid. The ZNE negative control further shows that improvement toward a reference is insufficient evidence unless signal retention, physicality, uncertainty, and a matched-cost control agree, matching concerns about artefactual ZNE improvement [5]. The combined experiment further supports a sep-

aration between estimator performance and mechanism validity: lower RMSE can coexist with rejected ZNE diagnostics and unresolved near-tie decisions.

6.2 Hypotheses and limitations

Possible causes of selector error include insufficient features, first-order ZNE, and the mismatch between synthetic training and shift channels; the experiment does not isolate these causes. The validity gate is conditional on its frozen thresholds, linear fit, chosen folds, and maximally mixed observable-floor model; it does not establish a general theorem about ZNE. All noisy values are simulator-derived. There is no IBM or other hardware execution, device drift, calibrated crosstalk, leakage, relaxation, or non-Markovian noise. Width is limited to six, the hardware-efficient ansatz and observables are fixed, and full assignment calibration scales exponentially. The same seed family and simulator implementation bound generalization. No ML selector is trained, and no post-holdout rule tuning is performed. The combined study covers only six preregistered cells, two budgets, and two allocation policies; it does not compare extrapolate-then-correct ordering or establish that readout and gate errors are separable on hardware. The scalable-readout mismatch study is limited to synthetic disjoint pair flips, one independent calibration approximation, widths at most six, and a single fixed decision margin; its negative result cannot quantify hardware crosstalk or drift. The application study covers only three deliberately controlled graph instances, five ideal-only candidates per instance, and $p = 1$. This curation is useful for known-reference falsification but is too small to support graph-family or scaling claims. Hardware evidence would require frozen candidates and budgets followed by device jobs that record calibration time, topology, transpilation, drift, leakage, and crosstalk; none of those effects is measured here.

A Reproducibility

```
python -m pytest
python -m src.verify_generated_artifacts
python -m src.verify_generated_artifacts --phase-f
python -m src.generate_paper_assets
tectonic paper/main.tex --outdir paper
```

Repeat-level rows remain under `results/raw`; processed summaries, manifests, run metadata, seeds, diagnostics, and execution counts remain separate and auditable. Tests check analytic references, statevector agreement, folding equivalence, shot accounting, deterministic seeds, manifest coverage, metric recomputation, and the holdout boundary.

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